BDS-0000 — Butler Design System
Version 1.0
Status: Baseline / Implementation Authority
Product: Butlerly

Purpose
BDS-0000 is the authoritative visual design-system specification for Butlerly. It converts the UX foundations into implementation-ready tokens, component rules, responsive behavior, theming, localization, accessibility, and Flutter constraints. UX documents define experience intent; BDS defines the reusable visual language and component contract.

1. Design Direction
Butlerly must feel calm, trustworthy, private, precise, modern, and efficient. The interface is content-first rather than decorative. One recognizable Butlerly identity is maintained across phone, tablet, and desktop while platform-native conventions are retained where they improve familiarity or accessibility.

Principles:
• Trust before delight.
• Quiet by default; reserve attention for decisions and exceptions.
• One clear primary action per decision area.
• Progressive disclosure for evidence, history, diagnostics, and advanced controls.
• Native, not identical, across platforms.
• Reversible actions favor Undo/history over unnecessary confirmation.
• Evidence remains near AI interpretation.
• User-entered and source data remain visually authoritative over AI proposals.

2. Design-System Layers
Foundation → primitives → controls → patterns → product compositions.
Foundation includes color, typography, spacing, sizing, radius, borders, elevation, motion, iconography, layout and content rules. Product code must consume semantic tokens rather than arbitrary values.

3. Token Contract
Naming: category.role.variant.state.
Examples: color.text.primary, color.surface.elevated, spacing.component.medium, radius.control.standard, motion.duration.fast, typography.body.standard.

Base spacing scale (dp): 4, 8, 12, 16, 24, 32, 48, 64.
Semantic aliases: micro=4, compact=8, small=12, standard=16, section=24, large=32, major=48, structural=64.

Radius: small=6, standard=10, large=16, full=999.
Border: hairline=1 logical pixel where supported; standard=1dp; emphasized=2dp.

Minimum touch target: 44×44dp; prefer 48×48dp for primary touch controls.

4. Color System
Butlerly’s primary brand identity is black / near-black, dark red / burgundy, and white / off-white. Neutral grays provide hierarchy and surfaces. Semantic colors are restrained and communicate meaning rather than decoration. Dark mode is the default visual direction and must not be produced by simple inversion.

Dark theme — authoritative V1 palette:
background #0B0B0D
surface #141418
elevatedSurface #1A1A1E
textPrimary #FFFFFF
textSecondary #A1A1A6
textTertiary #666E73
border #2A2A31
brand #B42333
brandDark #7A1825
interactive #E03A3E
selection #3A151B
success #2E9D64
warning #D6A84B
error #E06464
info #5A9BD5

Light theme — derived companion palette:
background #F7F7F8
surface #FFFFFF
elevatedSurface #FFFFFF
textPrimary #171719
textSecondary #5E5E64
textTertiary #7A7A80
border #DDDDE2
brand #A51F2E
brandDark #741722
interactive #B42333
selection #F5E1E4
success #2F855A
warning #9A6A16
error #B83A3A
info #3278A8

Color roles:
• Black / near-black is the dominant dark-theme foundation.

- White / off-white carries primary content and contrast.
- Dark red / burgundy is Butlerly's distinctive brand color.
- Interactive red is used for selected navigation, primary actions, focus emphasis, and other approved interactive states where contrast permits.
- Brand red and semantic error/destructive red are separate token roles even when visually related; components must consume semantic tokens rather than infer meaning from hue.
- Neutral gray is used for surfaces, borders, secondary text, disabled states, and hierarchy.
- Green is reserved primarily for success, positive status, and confirmed positive financial meaning.
- Amber is reserved primarily for warnings and attention states.
- Blue is informational/secondary only. It is not Butlerly's primary brand or interaction color.

Financial color rules:
- Red must not automatically mean expense. Because red is part of Butlerly's brand and interaction language, an expense is primarily communicated through its minus sign, label, amount context, iconography, or structure.
- Positive/negative financial meaning never depends on green/red alone.
- Income may use semantic positive green when useful, but must remain understandable without color.
- Destructive actions use the semantic error/destructive role, not the brand token merely because both are red-family colors.

Usage rules:
- Accent/brand color indicates identity or interaction, not decoration.
- Avoid rainbow dashboards. Category colors may be used selectively in charts where differentiation is necessary, but must not compete with the Butlerly black/red/white identity and must have non-color labels or legends.
- AI may use a subtle semantic indicator but never a theatrical gradient/glow as authority.
- Security and destructive colors are reserved for their semantic purpose.
- Generated UI reference images are subordinate to these semantic tokens. If an image contains a conflicting hex value, implement the BDS token while preserving the intended hierarchy and composition.
- All combinations must satisfy UX-0009 accessibility contrast requirements.

5. Typography
Default to the platform system font through Flutter platform-aware typography. Do not ship a custom primary font in V1 unless separately approved.

Roles (reference sizes before user scaling):
Display 32/40, semibold — rare hero/empty-state use.
Title Large 28/34, semibold — page title.
Title 22/28, semibold — section/major card title.
Heading 18/24, semibold — subsection.
Body 16/24, regular — standard reading.

Body Strong 16/24, semibold — emphasis.
Callout 15/21, regular — guidance.
Label 14/20, medium — controls/metadata labels.
Caption 12/17, regular — timestamps/provenance.
Monospaced Data 13/18 — identifiers/diagnostics only.

Use tabular numerals for aligned amounts when supported. Text scaling must reflow rather than clip. Critical financial digits may not be silently truncated.

6. Responsive Layout

Phone: <600dp logical width. Single primary column; bottom navigation; focused full-screen tasks; prominent capture/action affordance.
Tablet: 600–1023dp. Adaptive navigation rail/sidebar; list-detail and two-pane patterns where useful.
Desktop/Large: ≥1024dp. Persistent sidebar, global toolbar, multi-pane workspaces and tables where information density benefits.

These are layout breakpoints, not device identity checks. Window width determines composition. Preserve logical selection, scroll/filter state, and task progress when crossing breakpoints.

Content gutters: phone 16dp; tablet 24dp; desktop 32dp. Long-form readable content targets ~640–760dp maximum width. Data/table workspaces may expand to available width.

7. Navigation

Navigation destinations are defined by the authoritative product/UX information architecture; BDS does not introduce destinations. Phone uses bottom navigation when destination count permits. Tablet uses adaptive rail/sidebar. Desktop uses persistent sidebar plus toolbar. Stable destinations never reorder based on usage. Badges communicate actionable counts only.

8. Core Controls

Buttons: Primary, Secondary, Tertiary, Destructive, Icon. One primary button per decision area. Labels begin with verbs where practical. Loading preserves dimensions and blocks duplicate submission. Disabled state must remain readable.

Inputs: text, amount, date/time, search, select/menu, segmented selection, toggle, checkbox, radio, file and supported voice input. Visible labels are required; placeholder is never the only label. Validation occurs at useful moments and user input survives failures.

Cards group a coherent decision, summary, or preview; do not card-wrap every section. Sheets support compact contextual decisions. Dialogs are reserved for focused confirmation/authentication. Popovers and menus contain secondary actions, not hidden critical decisions.

9. Interaction States
Every applicable interactive component defines: default, hover, focus, pressed, selected, disabled, loading, success, warning, error and read-only. Focus remains visible in every theme and selected state. Hover is enhancement only; no required action depends on hover.

10. Lists and Data
Phone favors list rows with a clear primary target and limited trailing actions. Desktop/tablet may use tables for comparison, review and bulk operations. Tables require persistent column meaning, accessible headers, keyboard navigation, visible sort/filter state, tabular amount alignment and explicit selection for batch actions. Desktop may support Comfortable and Compact density; phone remains touch optimized.

11. Status, Empty and Error Patterns
Status vocabulary includes success, warning, error, offline, processing, pending changes, sync delayed, security alert, progress and Undo. Routine success and normal local/offline operation remain quiet.

Every empty state answers: what this area is, why it is empty, and what the user can do next. Every recoverable error states: what failed, what was preserved, whether data left the device when relevant, what remains usable, the next action, and whether retry is safe.

12. AI Presentation
AI Proposal: proposed value + provenance/evidence + material confidence + Accept/Edit/Dismiss.
AI Explanation: conclusion + evidence + uncertainty + action/recommendation + correction path.
Review Card: one meaningful exception + recommendation + evidence + confidence + one primary decision.
Assistant Answer: direct answer first + evidence + uncertainty + related records + next action.

AI content must never appear more authoritative than source-backed or user-corrected data. Corrected user values take precedence. AI confidence is textual/accessibly encoded. External processing boundaries and consent must be explicit when applicable.

13. Privacy and Security
Privacy language is specific and non-marketing. Consent has no preselected approval. Authentication prompts explain the protected action. Security alerts persist until understood/resolved when action is required. Sensitive content must not appear in notification previews or analytics by default.

No mandatory account is assumed for core operation. Components related to accounts, remote identity, device trust or synchronization must be capability-driven and must not make the local core experience dependent on sign-in.

14. Localization and Real Data

UI labels, system messages, navigation, help and product-authored content follow the selected application language and localization resources.

User-entered, imported, scanned/OCR and source-derived real data preserve their original language and script by default. Butlerly must not silently translate or rewrite source data merely because the UI language changes. Translation, if later offered, is an explicit derived view and does not replace the original value.

Dates, times, decimal separators, grouping and currency display follow locale where presentation semantics permit. Stored business meaning remains independent of display locale.

Currency amounts preserve original currency and original amount. When conversion is available, a normalized/reference amount may be shown separately, with USD as the default reference currency unless the user changes the preference. Converted values must identify the reference currency and must never replace the original transaction amount. Exchange-rate provenance/time must be available when conversion materially affects interpretation.

Layouts support text expansion and RTL where supported. Directional icons mirror appropriately. Do not embed translatable UI text in images or concatenate strings in ways that break grammar.

15. Accessibility

Accessibility is part of the component contract. Components support semantic names/roles/values, logical reading order, visible focus, keyboard/switch paths where applicable, large-text reflow, non-color status cues, reduced motion, sufficient contrast, accessible errors and minimum target sizes. UX-0009 is authoritative where requirements are stricter.

16. Motion and Haptics

Motion communicates relationship, progression or result; it is not decoration. Reference durations: fast 120ms, standard 200ms, deliberate 300ms. Prefer platform-native easing. Avoid looping decorative animation. Reduced-motion removes nonessential spatial movement. Haptics may confirm capture, save, error or destructive confirmation when platform appropriate. Sound is never required to understand state.

17. Platform Adaptation

Phone: one-handed priority, single-column progression, bottom navigation, sheets and full-screen capture/review.
Tablet: sidebar/list-detail, side-by-side source and record, touch/pointer/keyboard and adaptive panes.
Desktop: persistent navigation/toolbar, keyboard-first workflows, multi-pane layouts, tables, bulk actions and window restoration.

A platform-native component may replace a custom rendering when semantic token, state model, terminology, accessibility and outcome remain equivalent.

18. Flutter Implementation Contract

Create a centralized Butlerly design-system package/module. Raw colors, spacing and radii must not be scattered through feature widgets. Expose ThemeData/ColorScheme mappings plus Butlerly semantic extensions for roles not represented by standard Flutter theme APIs.

Recommended structure:
lib/design_system/tokens/
lib/design_system/theme/
lib/design_system/components/
lib/design_system/patterns/
lib/design_system/layout/

Use const values where appropriate, ThemeExtension for semantic theme tokens, MediaQuery/LayoutBuilder for adaptive composition, platform accessibility settings for text/reduced motion/high contrast, and Semantics/FocusTraversalGroup for accessible interaction. Feature code composes design-system components; it must not fork private visual variants without an approved BDS change.

19. Component Documentation Contract

Every production component records: component ID, purpose, anatomy, variants, states, behavior, content guidance, accessibility, platform adaptations, token usage, usage/misuse examples, engineering reference and tests.

IDs: C-CATEGORY-NNN. Examples: C-ACT-001 Primary Button; C-DATA-003 Record Row; C-AI-002 AI Explanation; C-SYS-004 Offline Banner.

20. Initial P0 Component Set

Foundations: semantic colors, typography, spacing/layout, sizing, radius/border/elevation, icons, motion, light/dark/system themes.
Controls: buttons, text/amount/date/search/select, toggle/checkbox/radio, menus, disclosure, progress.
Product primitives/patterns: app shell, navigation item, record row/detail section, capture controls, import progress/summary, Review Card, AI Proposal, AI Explanation, Source Preview, Consent Sheet, Sync/Capability Status, Offline/Error Banner, Undo Banner and Empty State.

P1 includes data table enhancements, duplicate/version comparison, advanced filters, batch toolbar, optional device-transfer/pairing components and diagnostics as product scope requires.

21. Governance

Design owns semantic intent, visual behavior, content and accessibility. Engineering owns implementation quality and platform integration. Product owns scope/prioritization. Privacy, security, AI autonomy and accessibility changes require cross-functional review.

Patch = documentation/visual correction without behavior change. Minor = backward-compatible token/state/component addition. Major = breaking semantic/API/interaction/token change. One-off feature variants require documented justification and either retirement or incorporation into BDS.

22. Design QA / Acceptance

A release UI passes BDS only when it verifies: correct tokens/components; supported widths; light/dark/system themes; text scaling/localization; keyboard/pointer/touch/screen reader; loading/empty/offline/degraded/success/error states; AI provenance/correction/consent; privacy boundaries; destructive action recovery; and no accidental exposure of private data.

23. Relationship to Butlerly Documentation

BDS-0000 is the design-system implementation authority. It is grounded in UX-0001 through UX-0012, especially UX-0007 Design System Foundations, UX-0009 Accessibility, UX-0010 Wireframes and UX-0011 Component Library. Product requirements remain authoritative for scope; architecture remains authoritative for system constraints; engineering standards remain authoritative for code quality. When documents conflict, higher-authority product/architecture decisions must not be overridden by a visual-design document.

24. Completion Criteria

BDS-0000 is complete for V1 baseline when designers can create high-fidelity screens without inventing foundational rules; engineers can implement a shared Flutter token/theme/component layer without guessing core values; QA can test visual, responsive, accessibility, localization, AI and privacy states; and future changes can be versioned without duplicating design rules across many documents.